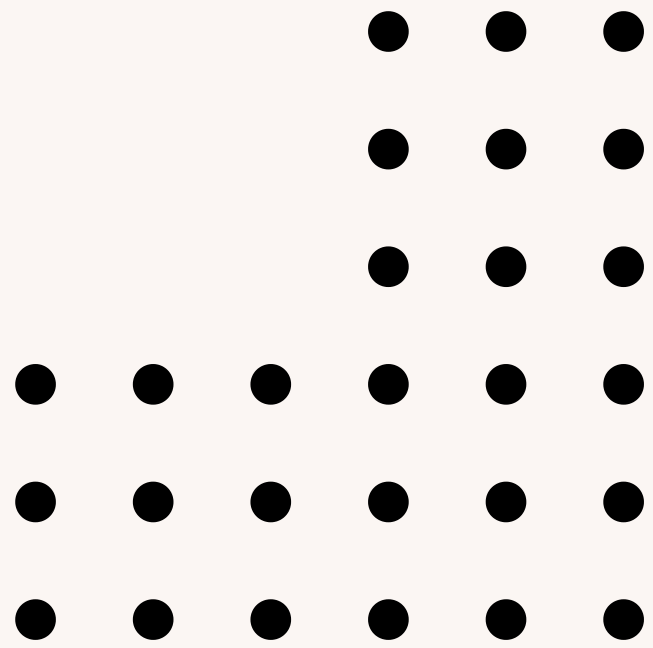




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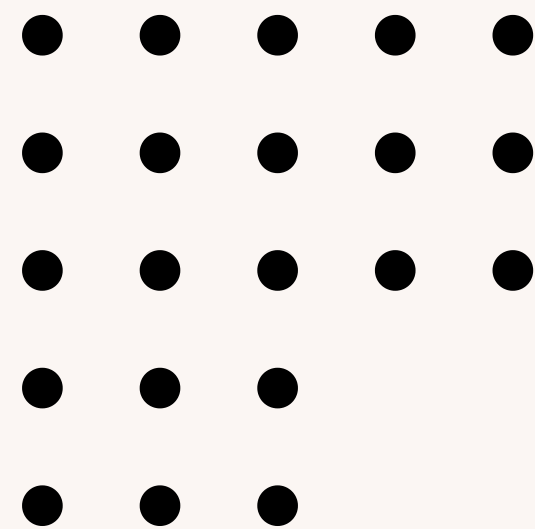
Material Library in GLSL (Nau)

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TOPICS

- ① The Lighting Problem
 - ①.1 The interaction between Light and Matter
 - ①.2 The fundamental role of the BRDF
- ② Physical Foundations
 - ②.1 Microfacet Theory
 - ②.2 The importance of Roughness
- ③ Analytical Reflectance Models
 - ③.1 The Cook-Torrance Model (D, F, G)
 - ③.2 Dielectrics vs. Conductors (Plastic vs. Metal)

TOPICS

④ Parameterization and Usability (PBR)

④.1 The Disney BRDF philosophy

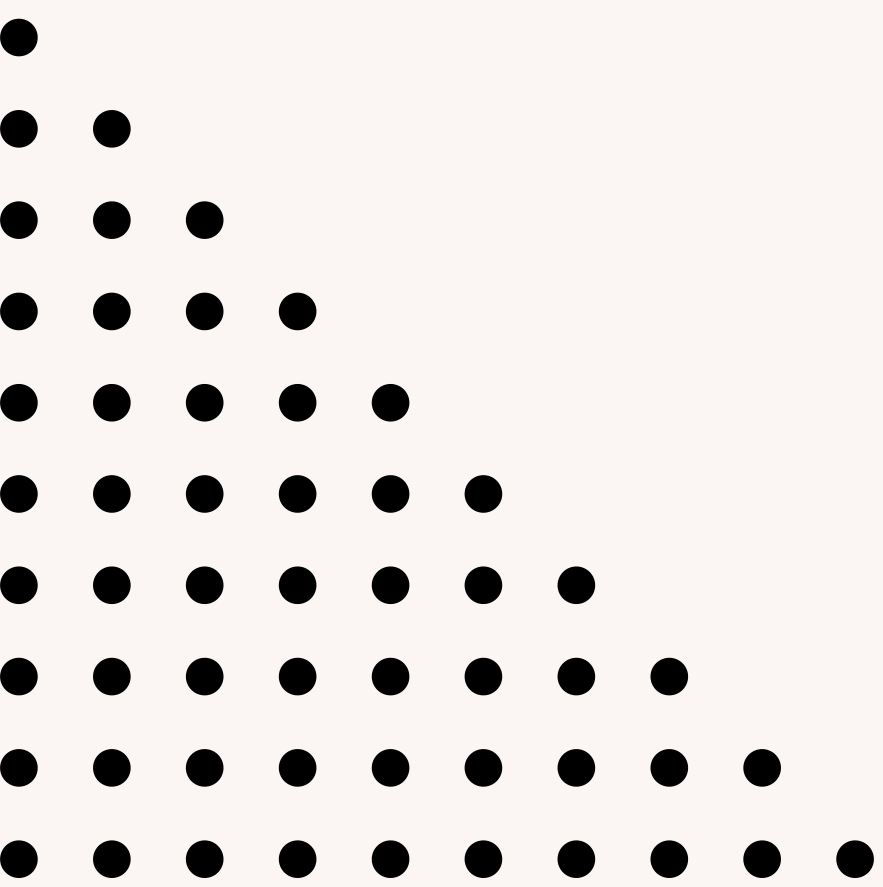
④.2 The Metallic-Roughness Workflow

⑤ Practical Application and Next Steps

⑤.1 Objectives until final delivery

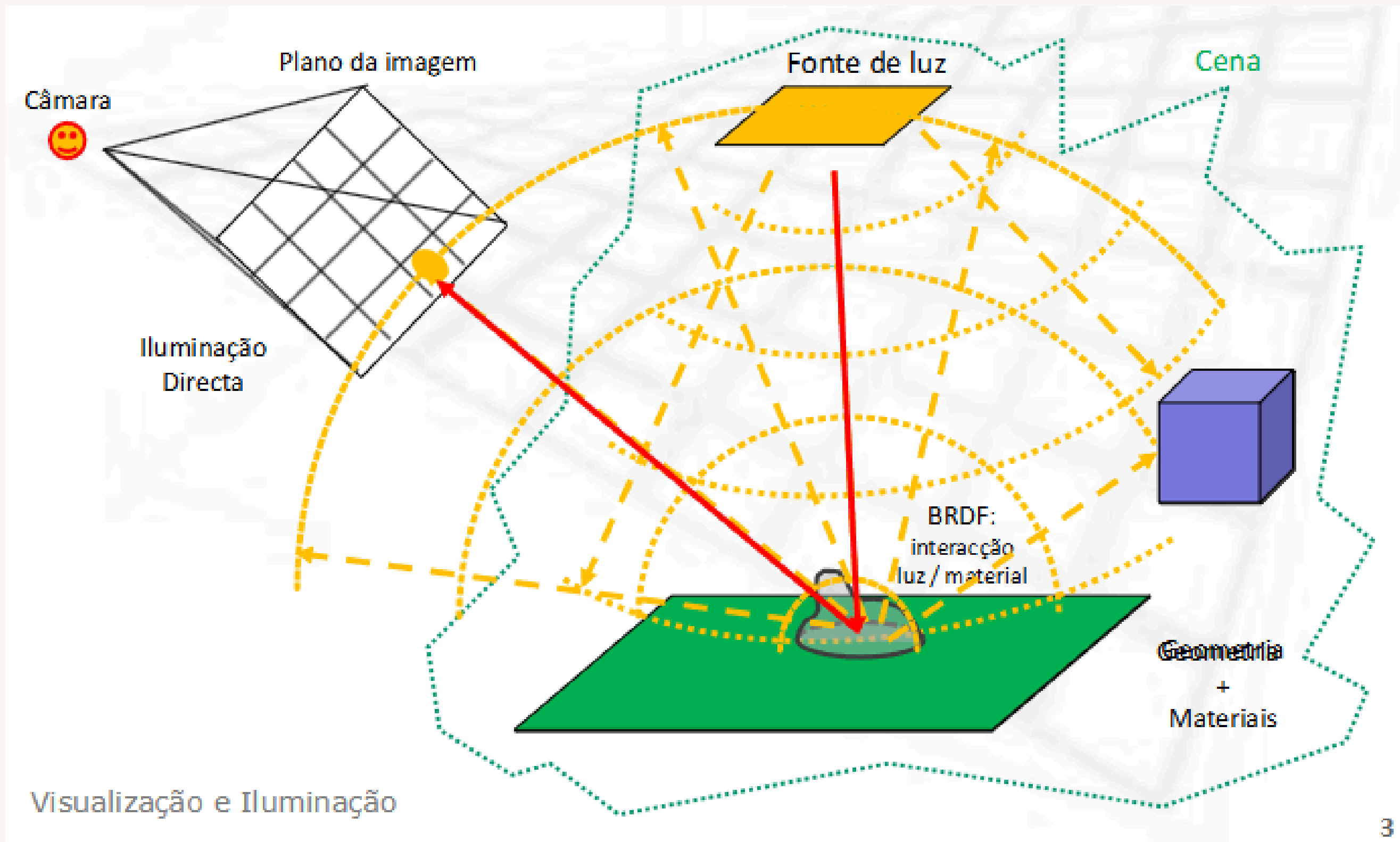
⑥ Conclusion

⑥.1 Bibliography



1. THE LIGHTING PROBLEM

1.1 THE INTERACTION BETWEEN LIGHT AND MATTER



1.2 THE FUNDAMENTAL ROLE OF THE BRDF

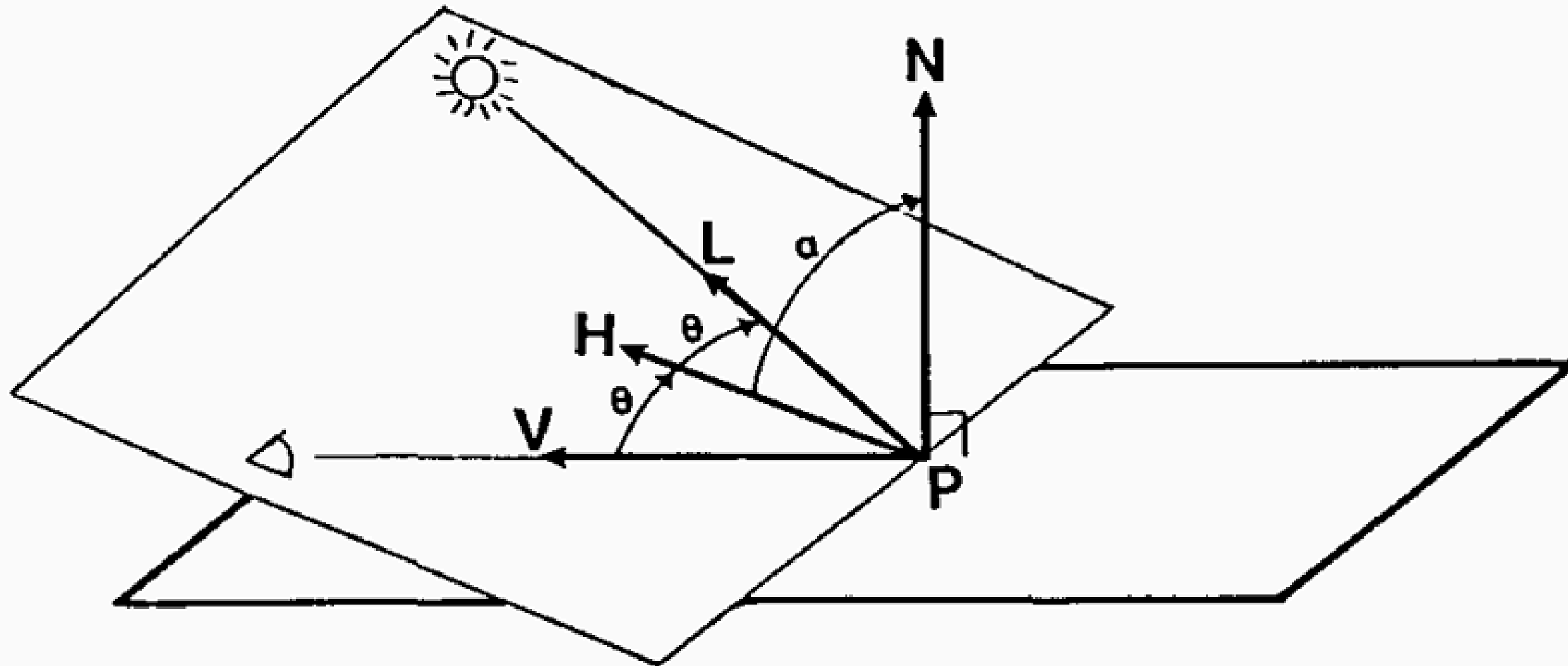
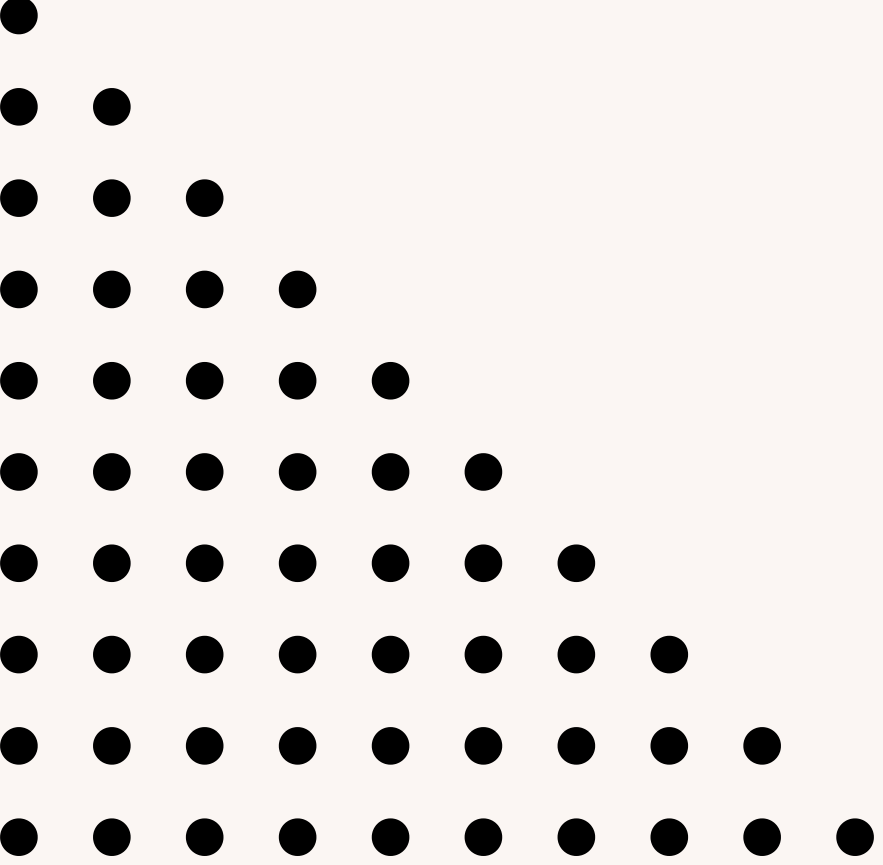


Figure 2. The geometry of reflection.



2. PHYSICAL FOUNDATIONS

2.1 MICROFACET THEORY

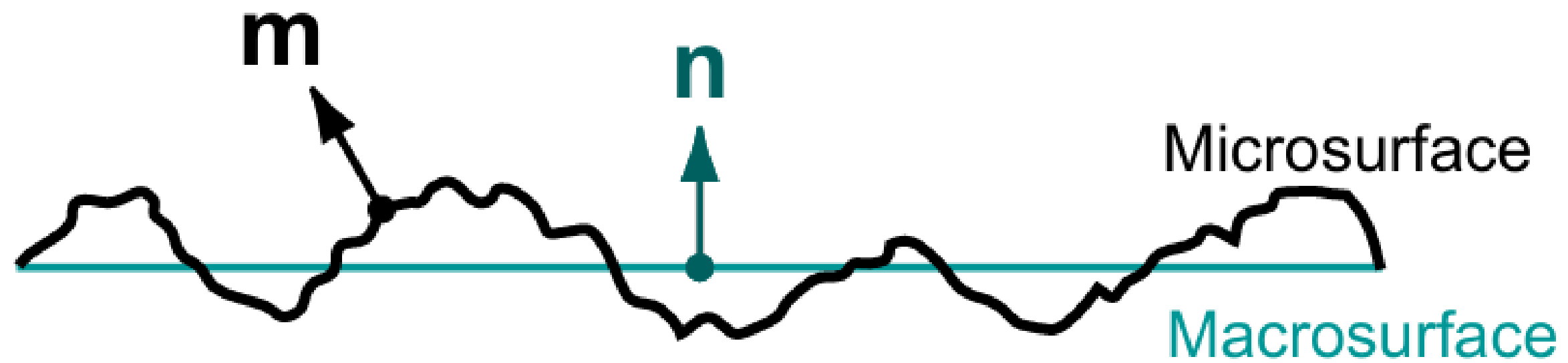
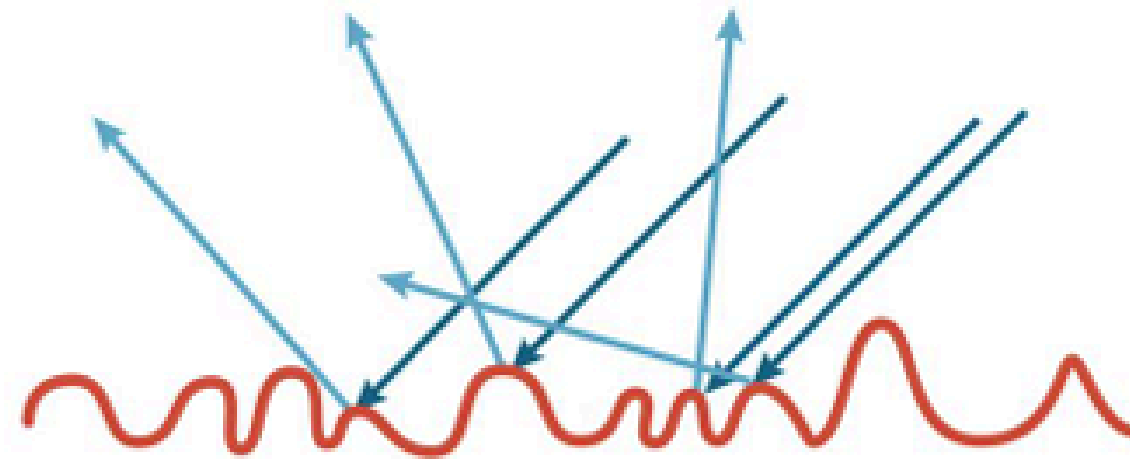
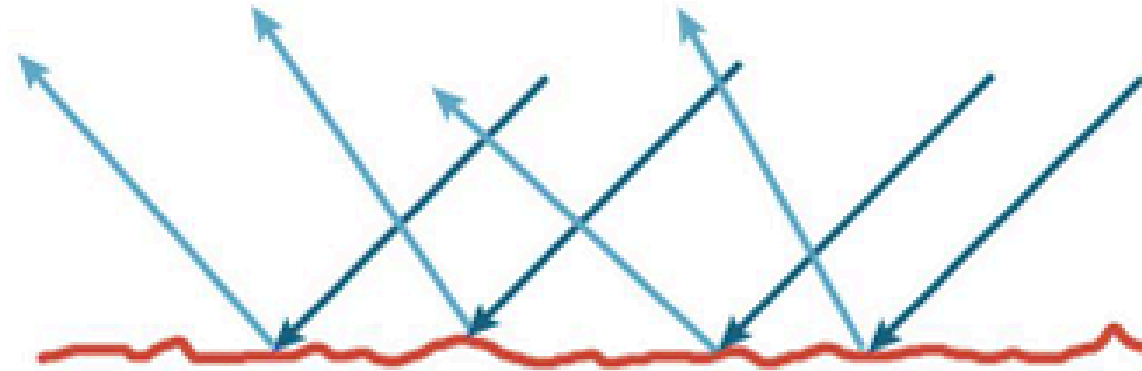
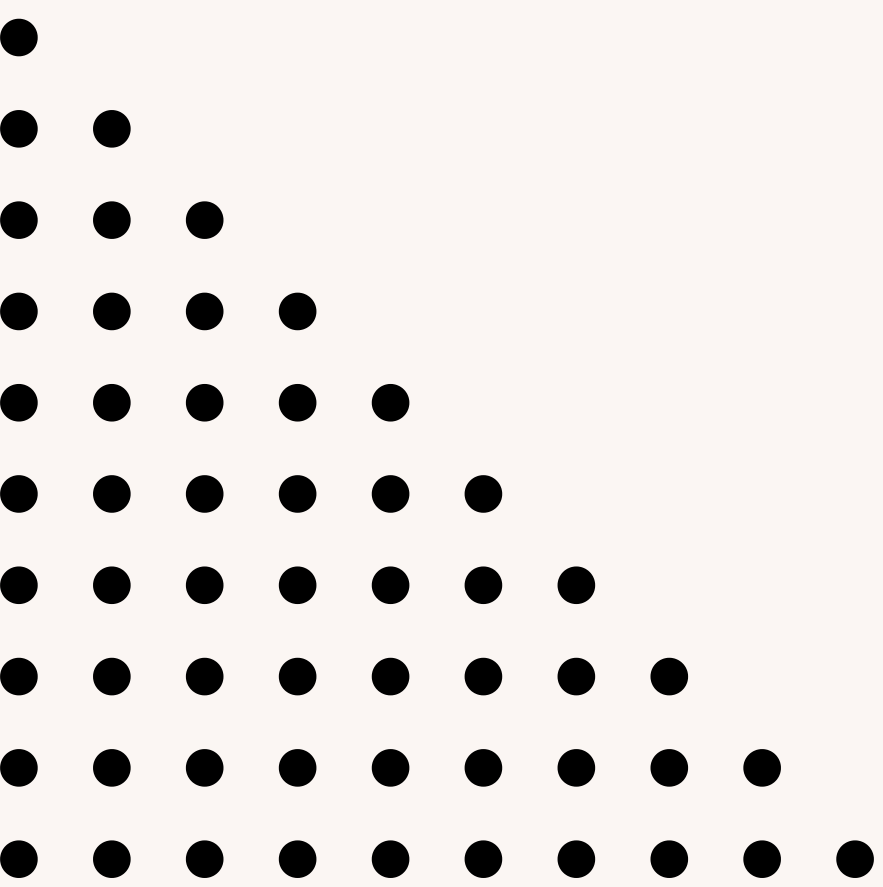


Figure 4: *Micro vs. macro surface.*

2.2 THE IMPORTANCE OF ROUGHNESS





3. ANALYTICAL REFLECTANCE MODELS

3.1 THE COOK-TORRANCE MODEL (D, F, G)

$$R = sR_s + dR_d, \quad \text{where } s+d=1.$$

$$R_s = \frac{F}{\pi} \frac{D}{(N \cdot L)} \frac{G}{(N \cdot V)}$$

F - Fresnel

Ratio of reflected vs. refracted light depending on the angle of incidence.

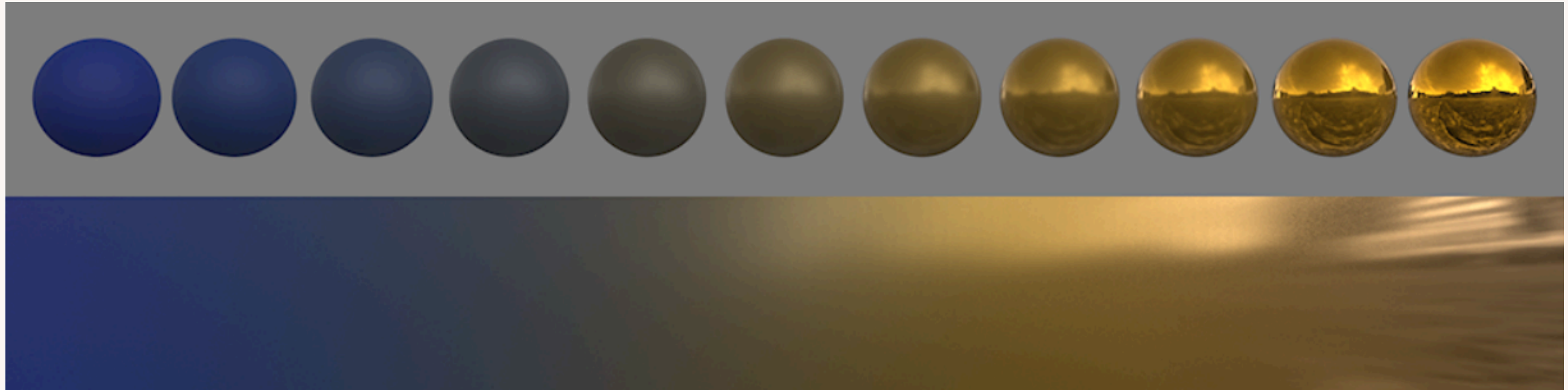
D - Normal Distribution

Fraction of microfacets perfectly aligned with the half-vector.

G - Geometry

Attenuation due to self-shadowing and masking between microfacets

3.2 DIELECTRICS VS. CONDUCTORS (PLASTIC VS. METAL)

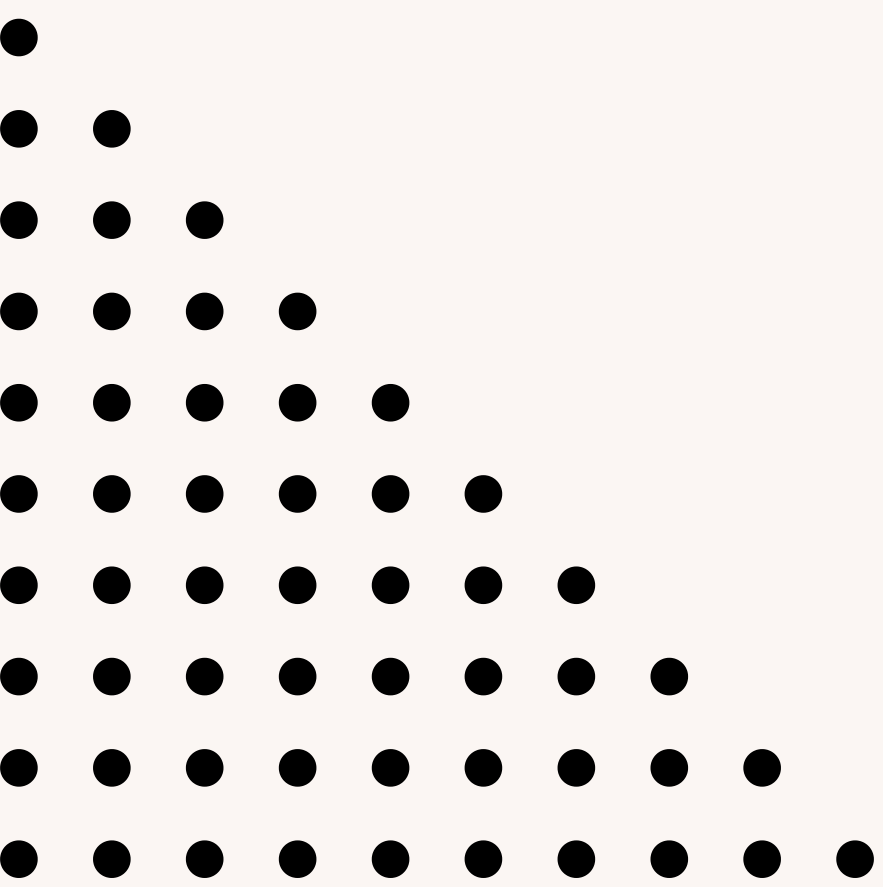
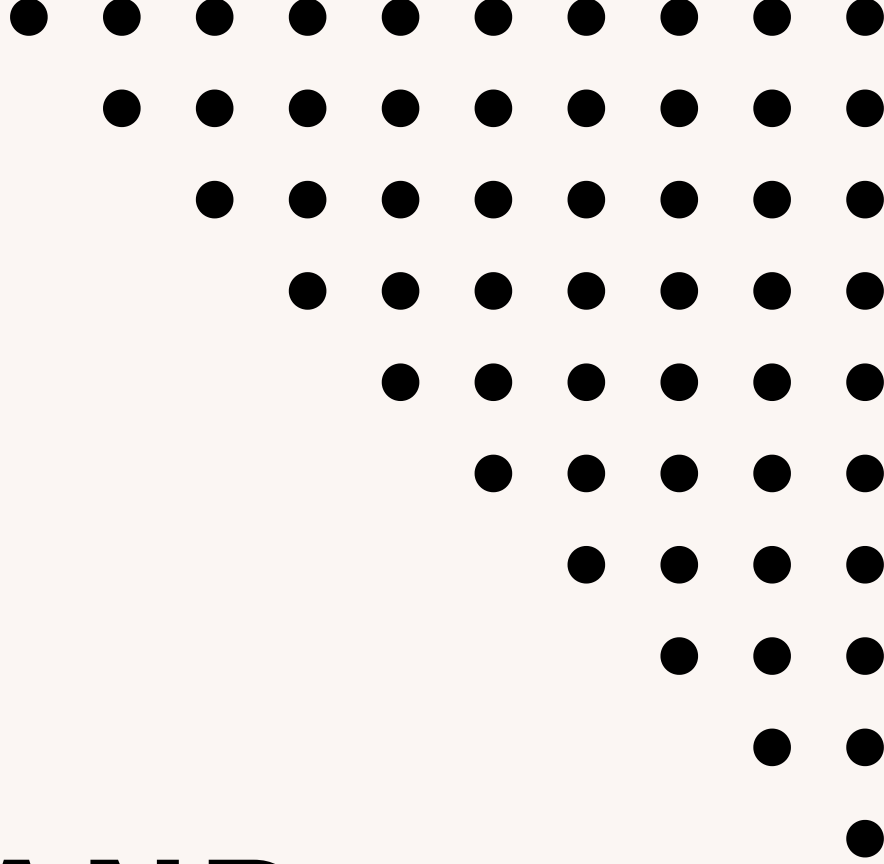


Dielectrics (Blue rubber)

- **Diffuse Color:** Visible
- **Reflection:** Colorless (only reflects the light's color)
- **Base Reflectance:** Very low (~4%)

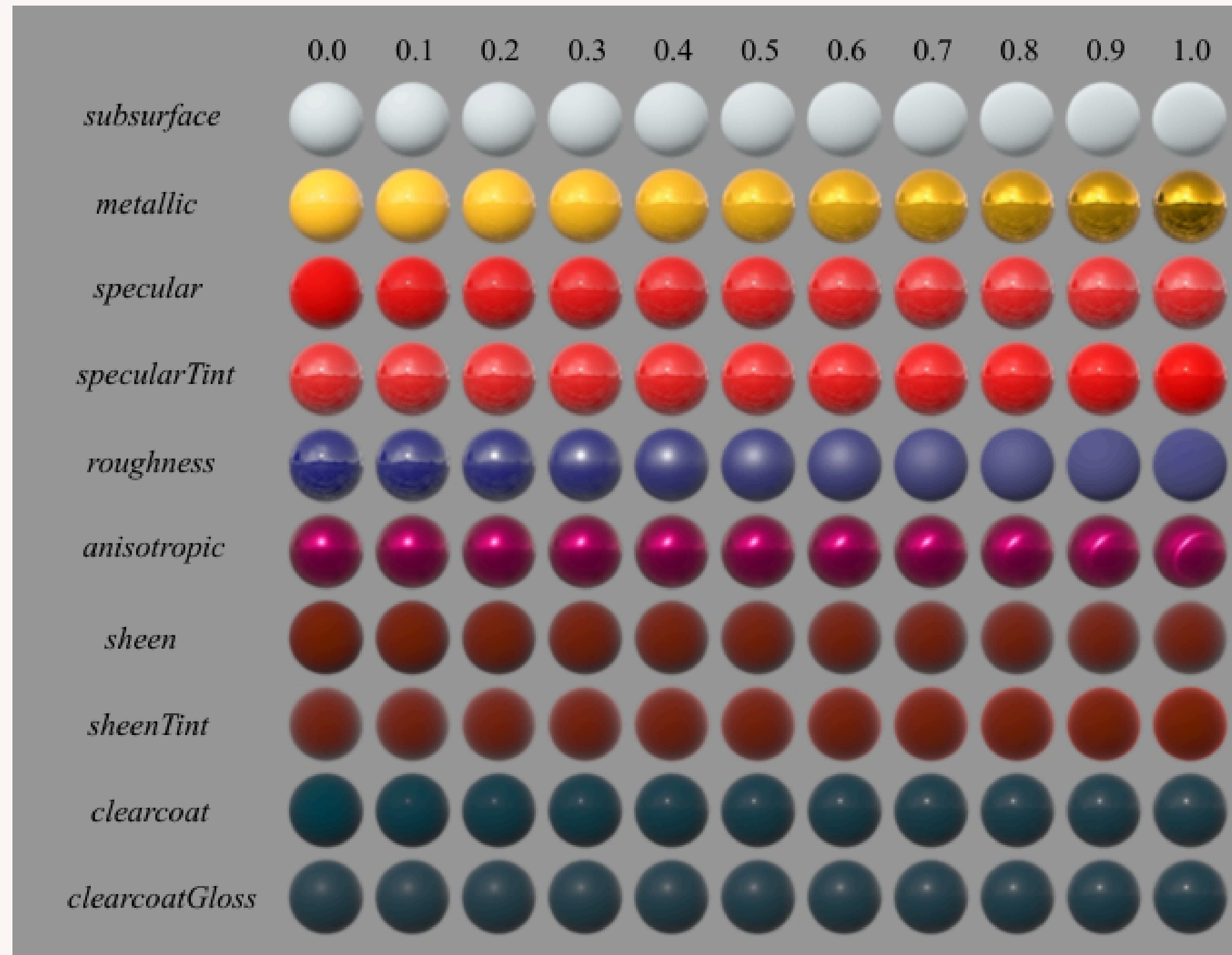
Conductors (Gold)

- **Diffuse Color:** None - it is absorbed
- **Reflection:** Colored (takes on the metal's color)
- **Base Reflectance:** Very high (50% to 100%)

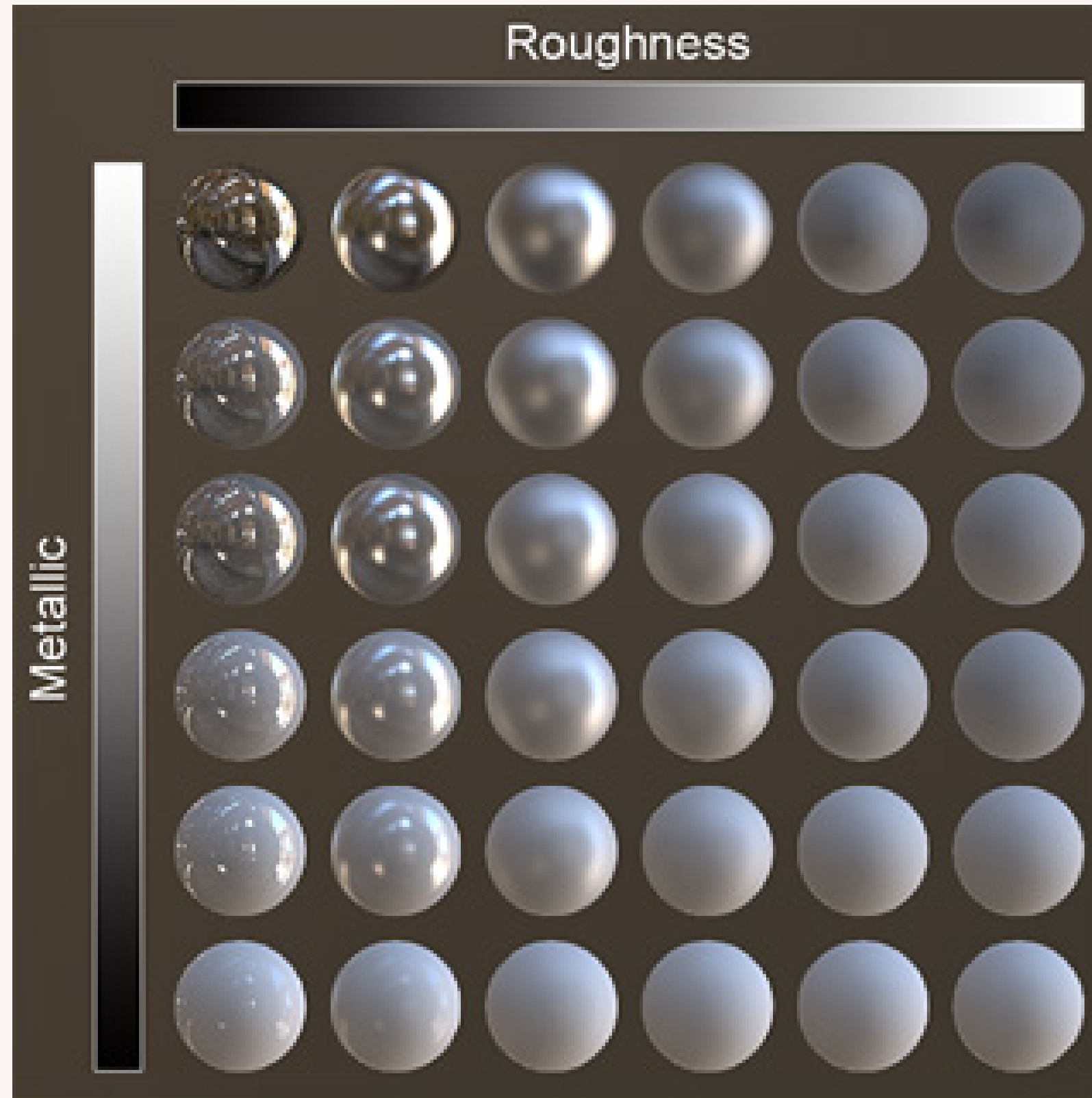


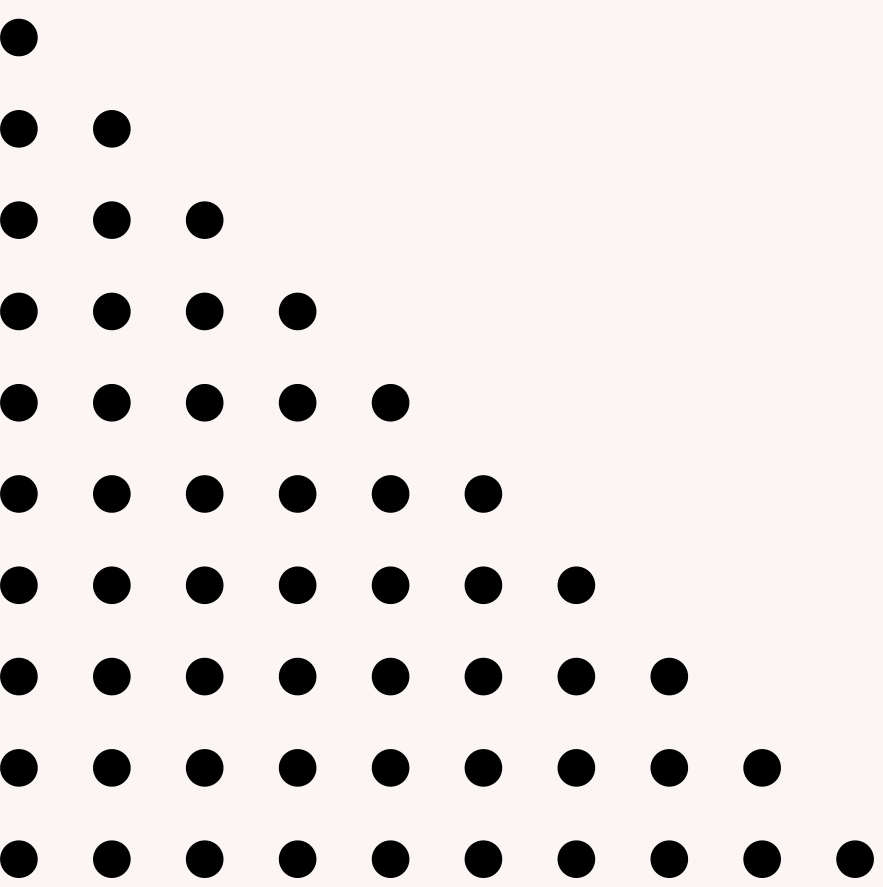
4. PARAMETERIZATION AND USABILITY (PBR)

4.1 THE DISNEY BRDF PHILOSOPHY



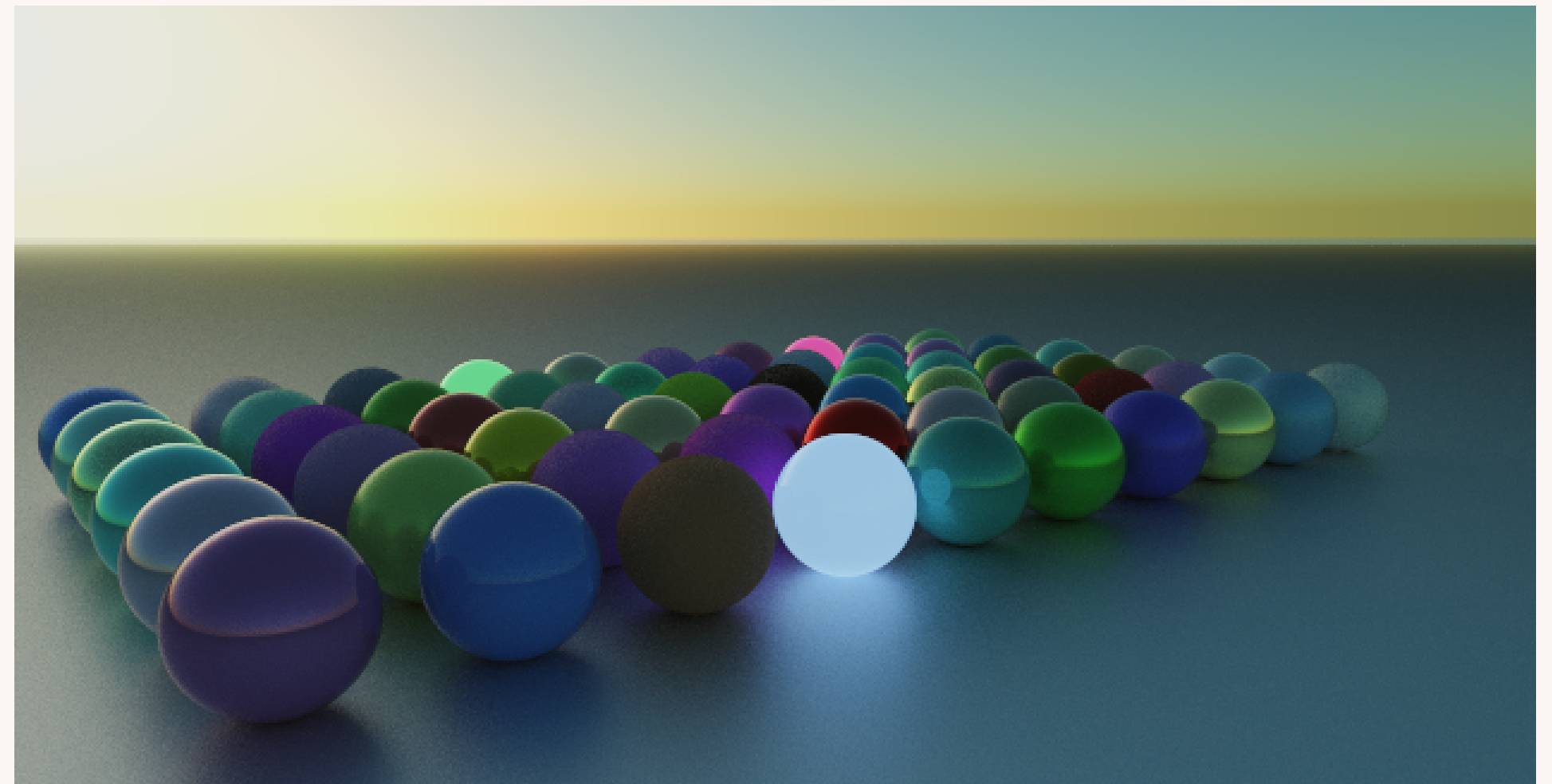
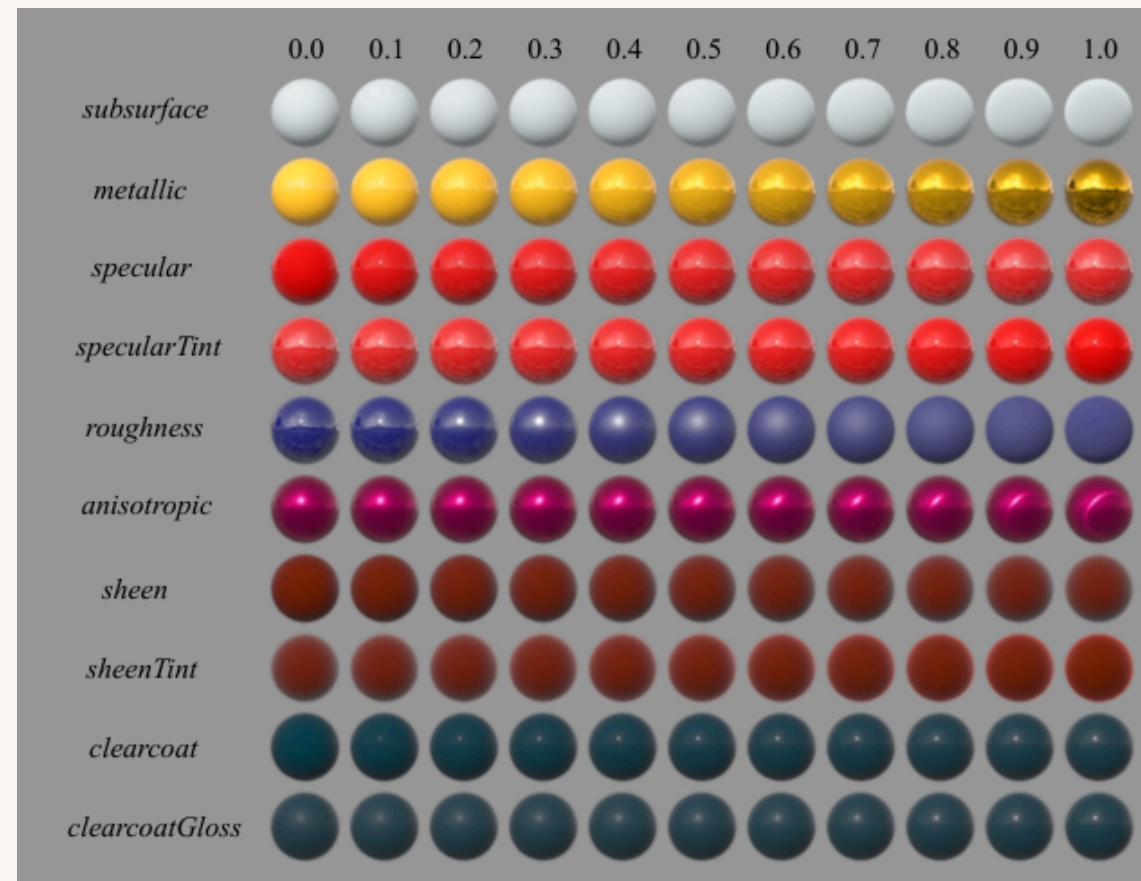
4.2 THE METALLIC-ROUGHNESS WORKFLOW

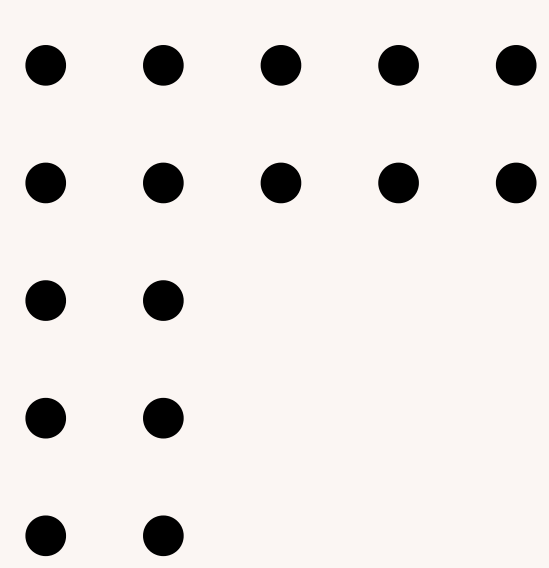




5. PRACTICAL APPLICATION AND NEXT STEPS

5.1 OBJECTIVES UNTIL FINAL DELIVERY





6. CONCLUSION

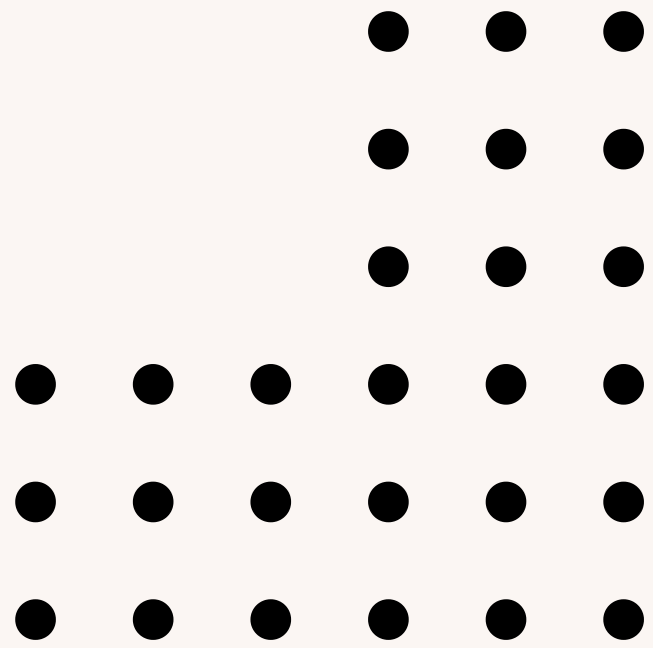
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Microfacet Models for Refraction through Rough Surfaces (Bruce Walter, Stephen R. Marschner, Hongsong Li & Kenneth E. Torrance)

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Measured values versus artistic input: Shading and material creation for film (Philip Engström, Tese de Licenciatura na Luleå University of Technology)



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